Data and preprocessing. … For Q2, , the same MCS data were combined with cantonal public holidays for the study period were scraped from feiertagskalender.ch with Playwright and parsed with BeautifulSoup, and permanent resident population per canton was downloaded from the Federal Statistical Office (BFS) to aggregate the holidays to Grossregion level. …

Analytical approach. … Q2 measures how much of the daily variation in spending is explained by weekday, month, year, COVID and public holidays, and then isolates the effect of regional holidays by comparing regions with and without a holiday on the same date (Section 3). …

3.1 Method

Holidays of all 26 cantons for 2019-2025 were scraped from feiertagskalender.ch (189 pages) with Playwright after checking robots.txt and the General Terms of Use (AGB), cached as HTML and parsed with BeautifulSoup. Full holidays count as 1, half-holidays as 0.5 and partial holidays by their stated population share. Weighting the cantons by their BFS population gives a daily holiday share (0-1) per Grossregion. This share was joined to the cleaned MCS data, from which codes 9999 had been removed. IQR outliers and the COVID period (16.03.2020-31.12.2022) were flagged rather than deleted. The resulting panel covers 2,557 days × 7 regions × 7 categories (125,293 observations).

Per category, log spending was regressed (OLS) on region, weekday, month, year, COVID and holiday share, adding one group at a time. Each group's R^2 gain is reported as a share of the within-region variation. A panel model with region and date fixed effects (Driscoll-Kraay standard errors) compares regions on the same date, so its holiday effect is identified only from the 301 dates on which holidays differ between regions. Effects are expressed as % change on a full holiday.

3.2 Results

Explained variation (Figure 3). Calendar effects explain most of the daily variation in retail but much less in hospitality. Within regions, they account for 91% of the variation in Retail: Food, beverage, tobacco, 78% in Retail: Other goods, 76% in Other and Transport services and 70% in Fuel stations, but only 60% in Food and beverage services and 48% in Accommodation. In Retail: Food, beverage, tobacco and Retail: Other goods, the weekday alone explains 65-74%: their Sunday index is 13-14, compared with 160-166 on Saturday, consistent with shops closing on Sundays and purchases moving to Saturday. Month effects are small except for December peaks in retail (index 122-133) and Transport services (125), a February peak in Transport (121) and a summer peak in Accommodation (July-August, 129-133). Year dummies explain most of the variation in Transport (66%) and Fuel stations (53%), possibly reflecting level

shifts during and after COVID. Holidays add a further 10-11 percentage points in the two retail categories and Other, and less than 1 point elsewhere.

Image: r2_by_category.png

Figure 3. Share of within-region daily variation in log spending explained by each calendar group, added in the order weekday, month, year, COVID, holiday (OLS per category, 2019-2025). Grey: variation not explained by calendar effects.

Regional holidays (Figure 4). Comparing regions on the same date, a full regional holiday lowers spending by 78% in Retail: Food, beverage, tobacco (95% CI -82 to -72), 73% in Retail: Other goods, 65% in Other and 52% in Transport services. The decline is smaller in Food and beverage services (-12%) and Fuel stations (-7%). The effect on Accommodation (-7%, 95% CI -16 to +3, p = 0.17) is not significant.

Image: holiday_effects.png

Figure 4. Change in daily scaled spending on a full public holiday by merchant category, 2019-2025 (% change, 95% confidence interval). Blue: all holidays (OLS with region, weekday, month, year and COVID controls). Red: regional holidays only (region and date fixed effects, same-date comparison).

3.3 Discussion and limitations

- **Retail.** The large drops match Swiss shop-closing rules: a holiday behaves like a Sunday. The same-date estimate is about 10 percentage points smaller than the all-holiday estimate, likely because residents can shop in neighbouring regions or move purchases to the days around a regional holiday, and because regional holidays rarely cover a whole region, so the full-holiday effect is estimated from days on which only part of the population is off.
- **Accommodation.** The positive all-holiday effect (+22%) disappears once regions are compared on the same date (-7%, not significant). A plausible reason is that national holidays coincide with long weekends and tourism, while a regional holiday does not.
- **Limitations.** Holidays are defined by canton, but spending is only available by Grossregion, which dilutes local holidays. The variance shares depend on the order in which variable groups are added, and year absorbs much of the COVID shock. Partial holidays with unknown coverage were set to zero.

Comparing regions on the same date removes everything that affects all of Switzerland on that day, but local events on regional holidays, such as fairs or carnival, may still be part of the estimated effect.